



State-of-the-art polymers of intrinsic microporosity for high-performance gas separation membranes

Yingge Wang, Bader S Ghanem, Yu Han and Ingo Pinnau

Solution-processible polymers of intrinsic microporosity (PIMs) are appealing materials for membrane applications due to their chemical and structural variety available from a wide range of PIM building blocks and significantly improved gas separation performance compared to conventional low-free-volume glassy polymers. This review highlights recent materials design developments in PIMs including: (i) benzotriptycene-based ladder PIMs, (ii) norbornyl-benzocyclobutene-based PIMs made by catalytic arene-norbornene annulation (CANAL), (iii) high-performance functionalized PIMs (iv) PIM-based thin-film composite membranes, and (v) PIM-based carbon molecular sieve (CMS) membranes. Significant advances in gas separation properties of new generation PIMs have set the recent 2015 H_2/N_2 , H_2/CH_4 , and O_2/N_2 upper bounds, 2018 CO_2/CH_4 mixed-gas upper bound, and 2019 pure-gas CO_2/N_2 and CO_2/CH_4 pure-gas upper bounds. Realizing the full potential of PIMs for future commercial use requires further improvements in (i) gas-pair selectivity, especially for challenging separations, (ii) stability against physical aging and plasticization, (iii) polymer scale-up, and (iv) reproducible membrane fabrication.

Address

Advanced Membranes and Porous Materials Center, Division of Physical Science and Engineering, King Abdullah University of Science and Technology, Al-Jazri Building, Thuwal 23955-6900, Saudi Arabia

Corresponding author: Pinnau, Ingo (ingo.pinnau@kaust.edu.sa)

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an annual compound growth rate of 6.0% [4]. However, currently used commercial low-free-volume glassy membrane materials, like cellulose acetate, polysulfone, poly(phenylene oxide), and conventional polyimides are limited by their low-to-moderate gas permeability and gas-pair selectivity to further advance membrane technology for existing and new gas separation applications.

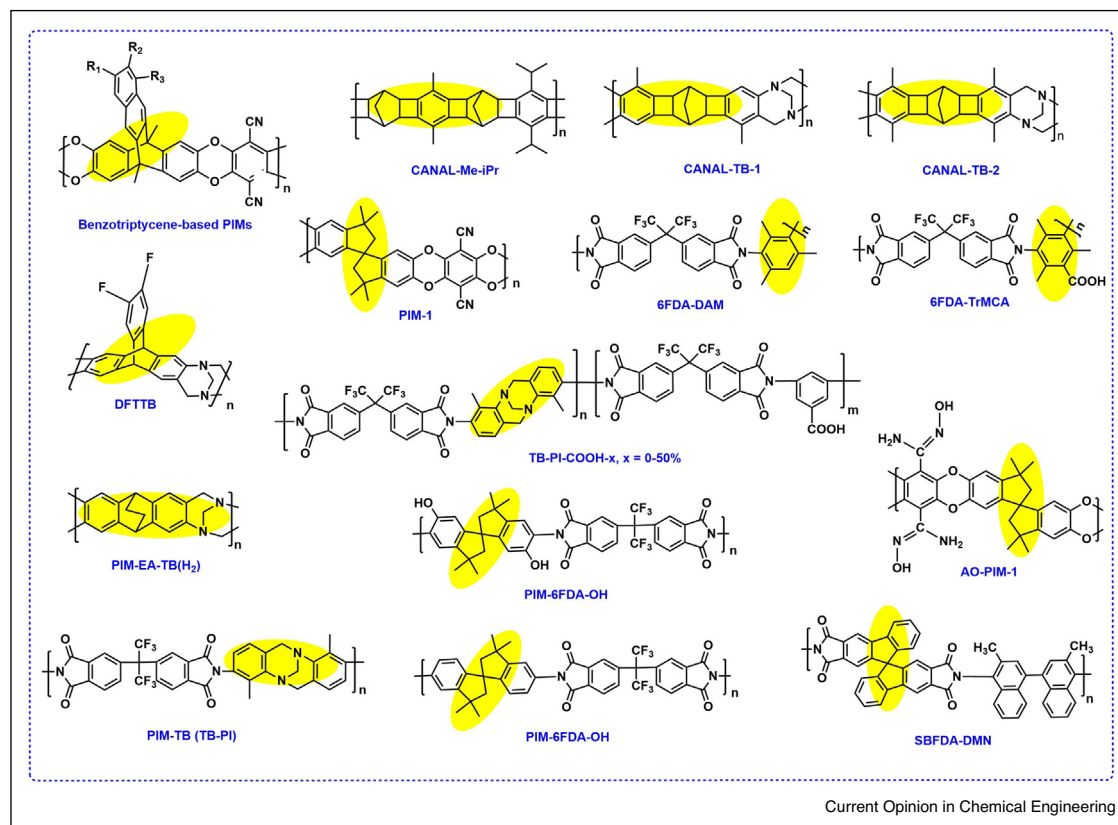
Polymers of intrinsic microporosity (PIMs) represent a unique class of microporous organic materials with great potential in membrane-based gas separation applications [5–12]. Incorporation of bulky, rigid contortion centers disrupts the polymer chain packing and results in high excess free volume, and hence, high gas permeabilities for PIMs. With an appropriate narrow bimodal distribution of micropores ($<20\text{ Å}$) and ultramicropores ($<7\text{ Å}$) in the polymer backbone, PIMs can transcend the traditional permeability/selectivity trade-offs and achieve enhancements in both gas permeability and selectivity [5–12]. The first ladder PIMs, designed with a contorted spirobisindane building block, defined the 2008 upper bounds for a number of important industrial separations [13]. Further developments of new PIM building blocks including triptycene (TRIP), ethanoanthracene (EA), and Tröger's base (TB) [14–21,22**] catalyzed several breakthroughs in gas separation membrane performance including the establishments of the 2015 H_2/N_2 , H_2/CH_4 , O_2/N_2 upper bounds [23], and 2019 pure-gas CO_2/N_2 and CO_2/CH_4 pure-gas upper bounds [24**]. In addition, incorporation of polar hydroxyl groups or carboxyl groups into the backbone structures of PIMs resulted in a number of high-performance functionalized PIMs defining the 2018 CO_2/CH_4 mixed-gas upper bound [8]. Recent studies also reported the significantly improved separation performance for PIMs tested at subambient temperatures ($-30 - 0\text{ °C}$) [22**,25**]. Furthermore, new developments in PIM-derived carbon molecular sieve membranes (CMS) and PIM-based mixed matrix membranes have demonstrated great potential in energy-intensive separations such as olefin/paraffin separations [8,9].

Introduction

Remarkable energy savings and substantial economic and environmental benefits provided by modern membrane technology have made it a viable alternative to replace energy-intensive conventional industrial unit operations such as distillation, absorption or adsorption processes [1–3]. Accordingly, the membrane industry has experienced significant growth in the past several decades with an estimated global market of up to 10 billion USD in 2021 at

Here, we highlight the research progress on newly developed high-performance benzotriptycene-based and norbornyl-benzocyclobutene-based ladder PIMs, functionalized PIMs, PIM-derived carbon molecular sieves (CMS), thin-film composite membranes, as well as gas transport properties of PIMs tested under subambient temperatures. The chemical structures and main properties of PIMs discussed in this review are presented in [Figure 1](#) and [Table 1](#), respectively.

Figure 1



Representative chemical structures of PIMs discussed in this review.

Novel high-performance ladder PIM materials

The inclusion of rigid, shape-persistent contortion centers, such as the 3D paddle-wheel triptycene motif, has produced a number of high-performance ladder PIMs which defined the 2015 upper bounds in O_2/N_2 , H_2/N_2 , and H_2/CH_4 separations [23]. However, these PIMs have not achieved much noteworthy progress in improving CO_2/N_2 and CO_2/CH_4 separations. Comesaña-Gándara *et al.* reported a number of ultrapermeable ladder PIMs prepared from the rigid and contorted benzotriptycene (BTrip) unit (Figure 1), which upon aging have shown great promise in CO_2/N_2 and CO_2/CH_4 separations [24^{••}]. For example, a fresh PIM-BTrip film (160 μm) exhibited an extraordinary CO_2 permeability of 215 00 Barrer (1 Barrer = $10^{-10} \text{ cm}^3(\text{STP}) \text{ cm cm}^{-2} \text{ s}^{-1} \text{ cmHg}^{-1}$) coupled with moderate CO_2/CH_4 selectivity of 12.7. After 130 days of aging, significantly improved sieving effects were observed as the CO_2/CH_4 selectivity nearly doubled (23.2) along with an approximately 40% loss in CO_2 permeability (21 500 versus 13 200 Barrer) [24^{••}]. Physical aging of PIM-BTrip continued with increasing CO_2/CH_4 selectivity and decreasing CO_2 permeability as the aging time prolonged to nearly two years. After 718 days of

aging, the PIM-BTrip film displayed a CO_2 and O_2 permeability of 3770 and 838 Barrer with CO_2/CH_4 and O_2/N_2 selectivity of 33.5 and 7.5, respectively (Table 1 and Figure 2) [24^{••}]. These remarkable gas permeation properties of PIM-BTrip and several other benzotriptycene ladder polymers including PIM-TMN-Trip, PIM-HMI-Trip, TFM-BTrip, and DTFM-BTrip defined the most up-to-date 2019 pure-gas CO_2/N_2 and CO_2/CH_4 upper bounds (Figure 2). However, this special class of PIMs has very limited solubility in organic solvents and requires using high boiling point solvents like quinoline to prepare film samples, which presents a great hurdle to process them for industrial scale-up and membrane fabrication. Additionally, high-pressure mixed-gas permeation properties of these materials are needed to further evaluate the full potential of these benzotriptycene-based ladder PIMs for natural gas sweetening or biogas treatment.

Another interesting new class of PIM materials with pure hydrocarbon backbone is based on norbornyl-benzocyclobutene ladder polymers prepared by catalytic arene-norbornene annulation (CANAL) [26[•],27,28[•],29[•]]. In

Table 1

Gas permeation properties of PIMs and their derivatives highlighted in this review

Polymer	Pure-gas permeability (Barrer)					Ideal selectivity		
	H ₂	N ₂	O ₂	CH ₄	CO ₂	H ₂ /N ₂	O ₂ /N ₂	CO ₂ /CH ₄
PIM-BTrip 160 μ m [24**]	12 100	1190	4330	1690	21 500	10.2	3.6	12.7
(130 days)	8440	522	2570	570	13 200	16.2	4.9	23.2
(718 days)	4820	112	838	113	3770	43.2	7.5	33.5
PIM-BTrip 64 μ m [24**]	9430	339	1800	412	9200	27.8	5.3	22.3
(120 days)	7180	200	1160	237	6040	35.8	5.8	25.5
(371 days)	7730	154	997	163	5150	50.2	6.5	31.6
PIM-TMN-Trip 166 μ m [24**]	18 800	3540	10,400	7250	52 800	5.3	2.9	7.3
(120 days)	15 300	1970	6620	3130	33 300	7.8	3.4	10.6
(426 days)	14 100	1100	4620	1440	20 400	12.8	4.2	14.2
PIM-HMI-Trip 135 μ m [24**]	16 600	2560	8540	4870	44 200	6.5	3.3	9.1
(120 days)	11 800	1440	5180	2150	26 900	8.2	3.6	12.5
(426 days)	11 000	804	3580	967	16 400	13.7	4.5	16.9
TFM-BTrip 176 μ m [24**]	13 600	1830	6210	2280	33 700	7.4	3.4	14.8
(123 days)	10 700	1090	4230	1250	22 100	9.8	3.9	17.7
(496 days)	9760	722	3260	792	15 600	13.5	4.5	19.7
DTFM-BTrip 112 μ m [24**]	14 700	3000	7770	4340	42 600	4.9	2.6	9.8
(119 days)	11 300	1800	5410	2150	29 000	6.3	3.0	13.5
(636 days)	10 200	741	3170	728	14 800	13.8	4.3	20.3
DM-BTrip 114 μ m [24**]	11 400	1020	3950	1570	22 000	11.3	3.9	14.0
(128 days)	9870	521	2640	599	12 200	18.9	5.1	20.4
DFTTB 63 μ m [22**]	5468	109	650	144	3146	50.2	6.0	21.8
(180 days)	2696	43	281	49	1005	62.7	6.5	20.5
CANAL-Me-iPr [28*]	1550	181	500	390	2340	8.6	2.8	6.0
CANAL-TB-1 [29*]	2760	97	463	121	1678	28.5	4.8	13.9
(300 days)	1163	39	204	53	749	29.8	5.2	14.1
CANAL-TB-2 [29*]	3608	162	747	205	2520	22.3	4.6	12.3
(300 days)	2452	110	528	129	1751	22.4	4.8	13.6
TB-PI [33**]	322	10	50	6.3	190	32.2	5.0	30.2
TB-PI-COOH-10 [33**]	271	6.9	35	3.8	138	39.3	5.1	36.3
TB-PI-COOH-20 [33**]	223	4.5	25	2.0	112	50.0	5.5	56.0
TB-PI-COOH-30 [33**]	180	2.5	17	1.0	76	72.0	6.8	76.0
TB-PI-COOH-40 [33**]	152	1.9	14	0.7	58	80.0	7.4	83
TB-PI-COOH-50 [33**]	138	1.7	13.7	0.6	37	81.2	8.1	62
6FDA-TrMCA [34**]	193	5.52	26.4	3.2	144	35	4.8	45
TB-PI (pristine) [43**]	390	12	47	8	285	33	3.9	35
TB-CMS-550 [43**]	14,600	780	4100	475	16,050	19	5.3	34
TB-CMS-650 [43**]	6552	118	1092	68	4200	56	9.3	62
TB-CMS-800 [43**]	2500	38	342	12.5	1406	66	9.0	112
SBFDA-DMN (pristine) [44**]	–	226	850	326	4700	–	3.8	14
CMS-600 [44**]	–	110	663	77.5	2853	–	6.0	37
CMS-700 [44**]	–	8.75	85	5.7	236	–	9.8	41
CMS-800 [44**]	–	3.23	35	0.93	105	–	11	113
CMS-900 [44**]	–	1.6	20	0.27	60	–	13	222
CMS-950 [44**]	–	1.06	12	0.104	39	–	11	375
CMS-1000 [44**]	–	0.7	7.52	0.02	30	–	11	1475
(250 days)	–	0.48	5.62	0.013	22	–	11.7	1732

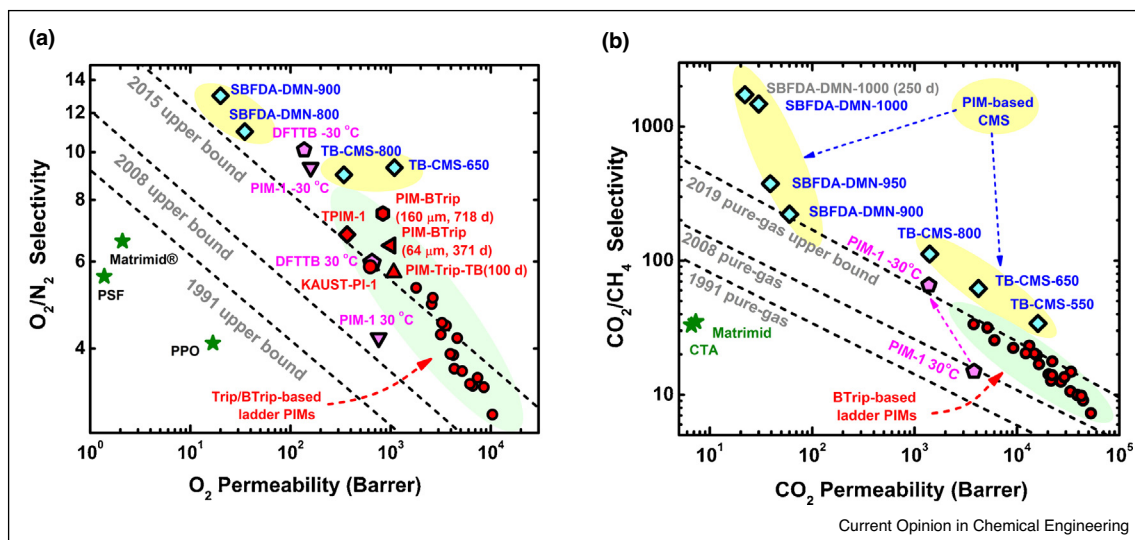
2019, Lai *et al.* reported three ladder PIMs based on CANAL polymerization with short alkyl substitutions. The methyl-substituted and isopropyl-substituted CANAL PIM (CANAL-Me-iPr) showed O₂ and CO₂ permeability of 500 and 2340 Barrer, respectively, but with O₂/N₂ and CO₂/CH₄ gas-pair selectivity of 2.8 and of 6.0, respectively [28*]. Ma *et al.* reported two new ladder PIMs prepared by introducing Tröger's base in CANAL polymer backbones. For example, CANAL-TB-1 displayed O₂ permeability of 463 and 204 Barrer coupled with O₂/N₂ selectivity of 4.8 and 5.2 for fresh and

300-day-aged samples, respectively, demonstrating great potential for air separation applications [29*].

Highlights on the progress of functionalized PIMs

Functionalized polymers with polar substituents, such as hydroxyl groups or carboxyl groups, are well known to exhibit better size-sieving properties and resistance to penetrant-induced plasticization due to polymer chain tightening effects induced by inter-chain or intra-chain interactions through either hydrogen bonding or π - π

Figure 2



Gas separation performance of representative PIMs and PIM-based CMS membranes; (a) O_2/N_2 separation; (b) CO_2/CH_4 separation. Light green highlighted areas include data from Trip or B-Trip-based PIMs. Yellow highlighted area includes data from PIM-based CMS membranes. Closed pink labels highlight the subambient temperature data of ladder PIMs.

interactions [8,9,30–32]. A number of hydroxyl-functionalized PIMs and carboxylic functionalized co-polyimides reported in the literature defined the 2018 CO_2/CH_4 mixed-gas upper bound [8]. Recent notable progress in this research direction includes the development of carboxyl-functionalized PIM-PI copolymers and homopolymers as discussed below [9].

Wang *et al.* [33**] reported a series of high-performance 6FDA-TB-derived copolyimides containing up to 50% carboxylic acid groups in the repeat unit. Increasing the carboxyl content and the tertiary amine moieties of TB units in the polymer backbone led to physical crosslinking with stronger intra-chain and inter-chain interactions, therefore, enhanced size-sieving properties and greatly improved gas separation performance and plastization resistance. As a result, the copolyimides showed gas permeation properties on or above 2008 H_2/N_2 , O_2/N_2 , and CO_2/CH_4 upper bounds. For example, the copolyimide containing 20, 30, and 40% carboxyl groups exhibited pure-gas CO_2 permeability of 112, 76, and 58 Barrer coupled with remarkable CO_2/CH_4 selectivities of 56, 76, and 83 respectively [33**]. More importantly, TB-PI-COOH-20 copolymer demonstrated strong plasticization resistance and stable separation performance up to 32 bar under aggressive 1:1 $CO_2:CH_4$ mixed-gas permeation conditions tested at 25°C [33**]. In fact, the excellent combination of mixed-gas CO_2 permeability (~80 Barrer) and $CO_2:CH_4$ selectivity (46) tested at 10 bar CO_2 partial pressure positioned the copolyimide containing

20% carboxyl groups (TB-PI-COOH-20) above the 2018 CO_2/CH_4 mixed-gas upper bound [33**].

Abdulhamid *et al.* [34**] synthesized a novel PIM-PI homopolymer containing carboxyl groups (6FDA-TrMCA) using 6FDA and 3,5-diamino-2,4,6-trimethylbenzoic acid (TrMCA). Compared to its non-functionalized structural analogue 6FDA-TrMPD (6FDA-DAM), 6FDA-TrMCA had lower BET surface area (260 versus $450\text{ m}^2\text{ g}^{-1}$) and higher density due to tighter chain packing from hydrogen bonding through functional carboxyl groups, hence, exhibited lower gas permeability but much improved size-sieving properties, that is, higher gas-pair selectivity. The pure-gas CO_2 permeability of 6FDA-TrMCA was lowered by more than three-fold (144 versus 498 Barrer) with an enhanced CO_2/CH_4 selectivity from 22 to 45 [34**]. Similar to carboxyl-functionalized TB-based copolyimides, the 6FDA-TrMCA homopolymer achieved superior plasticization resistance up to 30 bar when tested using an aggressive 50:50 vol% CO_2/CH_4 mixture at 35°C. At 12 bar of CO_2 partial pressure, 6FDA-TrMCA displayed excellent mixed-gas CO_2/CH_4 performance, located on the 2018 CO_2/CH_4 mixed-gas upper bound, with a mixed-gas CO_2 permeability of 98 Barrer coupled with a CO_2/CH_4 selectivity of 38 [34**]. Combined with several hydroxyl-functionalized PIMs, incorporating various polar functional groups into the PIM-PI backbones is a proven strategy to successfully prepare high-performance PIMs for CO_2/CH_4 separations.

Besides hydroxyl- and carboxyl groups, sulfonyl groups, tetrazole, and/or amidoxime can be of interest to design advanced functionalized PIMs. For example, amidoxime-functionalized PIM-1, referred to as AO-PIM-1, was identified by Yi *et al.* [35**] as an outstanding candidate for sour gas treatment due to its exceptional separation performance for H₂S/CH₄ and CO₂/CH₄ separation. In a ternary mixture containing 20 mol% H₂S: 20 mol% CO₂: 60 mol% CH₄ and at high feed pressure up to 77 bar, AO-PIM-1 exhibited an ultrahigh mixed-gas H₂S permeability of >4000 Barrer combined with a H₂S/CH₄ selectivity of ~75 — an unmatched performance far exceeding any other material reported to date [35**]. It is worth mentioning that the undesirable penetrant-induced plasticization effects often observed for CO₂/CH₄ separation, paradoxically enhanced the H₂S/CH₄ mixture separation performance for AO-PIM-1.

Promise of PIM materials for low-temperature gas separation applications

Ma *et al.* [22**] recently reported interesting low-temperature permeation results for a ladder Trip-TB made from 2,3-difluoro-2,6(7)-tritycenediamine (DFTTB). At 35 °C, the O₂ permeability of a DFTTB film aged for 180 days (Table 1) was 281 Barrer coupled with an O₂/N₂ selectivity of 6.5; these properties were similar to triptycene-Tröger's base ladder polymers [17,21]. However, DFTTB exhibited a remarkable boost in performance for O₂/N₂, H₂/N₂, CO₂/N₂, and CO₂/CH₄ when tested at subambient temperatures. For example, at −30 °C DFTTB displayed an O₂/N₂ selectivity of 10.1 coupled with O₂ permeability of 137 Barrer and H₂/N₂ selectivity of 152 combined with H₂ permeability of 1532, far exceeding 2015 O₂/N₂ (Figure 2a) and H₂/N₂ upper bounds [22**]. The low-temperature transport data provided in this work demonstrated significant commercial potential of high-performance PIMs for low-temperature air separation, H₂ recovery, and natural gas treatment applications.

A similar improvement in gas separation performance at sub ambient temperature has also been recently reported for PIM-1 by Ji *et al.* [25**]. As the testing temperature decreased, the gas-pair selectivity was significantly improved with a concurrent drop in gas permeability. For example, the O₂ and CO₂ permeability of PIM-1 decreased from 759 to 159 Barrer and from 3798 to 1380 Barrer, respectively, by changing the testing temperature from 30 to −30 °C [25**]. However, this resulted in a significant increase in O₂/N₂ selectivity from 4.2 to 9.3 and in CO₂/CH₄ selectivity from 14.9 to 65.7 (Figure 2a and b) [25**]. Testing at subambient temperature also slowed down the aging effects for PIM-1 coupled with continued increase in CO₂/CH₄ selectivity. These two low temperature transport studies on highly permeable PIMs suggested a general trend of achieving high gas-pair selectivity at subambient temperatures and

great potential to obtain high-performance gas-separation properties.

PIM-derived high-performance CMS membranes

Compared to pristine polymeric membranes, carbon molecular sieve membranes typically show enhanced gas separation performance due to their superior size-sieving properties [36–38]. The gas separation performance of CMS membranes can be fine-tuned using proper selection of pristine carbon polymer precursors and pyrolysis conditions such as temperature, ramping rate, pyrolysis time, atmosphere, and so on. Extensive research efforts have been devoted to CMS membranes derived from a wide range of polymers including phenolic resins, polyimides such as Matrimid [36–38]. Compared to traditional low-free-volume glassy polymer precursors, the microporous nature of PIMs as the starting point for CMS formation offers a short-cut for CMS membranes to undergo structural changes through the thermal treatment. Thus, the inherent microporosity of pristine PIMs provides a new tool for structural formation of desirable permeable and ultra-selective submicroporous and ultra-microporous CMS membranes. However, only a few PIM-derived CMS membranes have been recently reported in the literature [8,9].

Early work on CMS membranes derived from PIM-1, PIM-6FDA-OH, and PIM-6FDA demonstrated great potential of PIM-based CMS for a number of challenging separations including CO₂/CH₄, C₂H₂/C₂H₆, and C₃H₆/C₃H₈ [39–41,42*]. For example, Salinas *et al.* [40] reported a C₂H₄ permeability of 1.3 Barrer and C₂H₄/C₂H₆ selectivity of 13 for a PIM-1-based CMS membrane pyrolyzed at 800 °C. CMS membranes derived from a hydroxyl-functionalized PIM-PI precursor (PIM-6FDA-OH) carbonized at 800 °C exhibited even better performance with a pure-gas ethylene permeability of 10.4 Barrer and C₂H₄/C₂H₆ selectivity of 17.5 [41]. Salinas *et al.* also reported CMS membranes made from PIM-6FDA, the hydroxyl-free analogue of PIM-6FDA-OH, which showed a lower pure-gas ethylene permeability of 3 Barrer coupled with an enhanced pure-gas C₂H₄/C₂H₆ selectivity of 25 [42*]. More importantly, the CMS membrane showed stable performance with a mixed-gas C₂H₄/C₂H₆ selectivity of 15.6 when tested in a 50:50 C₂H₄/C₂H₆ mixture at 20 bar [42*], which is the highest reported value reported in the literature under such challenging testing conditions.

Wang *et al.* [43**] reported CMS membranes derived from microporous 6FDA-Tröger's base polyimide (TB-PI) carbonized at different temperatures ranging from 550 to 800 °C. Ultrapervious CMS membranes were obtained at a pyrolysis temperature of 550 °C for 2 hours with a CO₂ permeability of 16 050 Barrer and CO₂/CH₄ selectivity of 34. At 800 °C, TB-CMS-800 showed excellent CO₂/CH₄ separation properties with a combination

of CO₂ permeability of 1406 Barrer and CO₂/CH₄ selectivity of 112 (Figure 2b) [43^{••}].

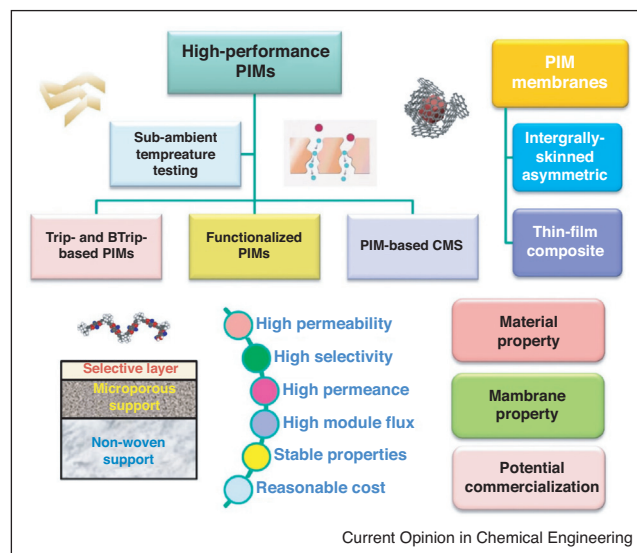
Hazazi *et al.* [44^{••}] prepared CMS membranes derived from a carbon-rich microporous polyimide, SBFDA-DMN, carbonized at various temperatures in the range of 550–1000 °C. Increasing the pyrolysis temperature induced significant structural changes in these CMS membranes, leading to development of ultra-selective CMS membranes. For example, the SBFDA-DMN-derived CMS membrane carbonized at 1000 °C demonstrated extraordinary molecular-sieving capacities with N₂/CH₄ and CO₂/CH₄ selectivity of 35 and 1475, respectively, far exceeding the 2019 polymer CO₂/CH₄ performance upper bound (Figure 2b) [44^{••}]. Based on these early results and the structural variety of PIM precursors, PIM-based CMS membranes have great potential for difficult gas separation applications, such as C₂H₄/C₂H₆, C₃H₆/C₃H₈, CO₂/CH₄, N₂/CH₄ and so on.

Transforming PIM materials into asymmetric and thin-film composite membranes

Within less than twenty years of active research, PIM materials have significantly outperformed conventional low-free-volume glassy polymers such as cellulose acetate, polysulfone, and polyimides in terms of their bulk gas separation performance measured using dense isotropic films with thicknesses >50 µm, which is far from industrial requirements for large-scale membrane fabrication where thicknesses of typically less than 1 µm are required. Successfully integrating PIMs into asymmetric or thin-film composite membranes has become the key to fully realize their full commercial potential (Figure 3). However, only limited successful approaches have been reported to date. Defect-free, integrally skinned asymmetric (ISA) PIM-1 hollow fiber membranes were first reported by Jue *et al.* [45[•]] using a triple orifice spinneret and a modified 1-butanol/THF sheath layer dual-bath spinning technique. After two-months of physical aging, the defect-free PIM-1 hollow fiber membrane with skin layer thickness of ~2.8 µm exhibited a high CO₂ permeance of 360 GPU combined with a CO₂/CH₄ selectivity of 23 that was higher than commonly reported for bulk films [45[•]]. PIM-based triple-layer composite hollow fiber membranes were prepared by Liang *et al.* [46] using a selective layer composed of PIM-1 and/or PIM-1 copolymer with β-cyclodextrin (PIM-CD), a cross-linked polydimethylsiloxane (PDMS) gutter layer, and a microporous polyacrylonitrile (PAN) support. The triple-layer PIM-1/PDMS/PAN composite hollow fiber membrane displayed CO₂ permeance of 403 GPU coupled with pure-gas CO₂/CH₄ selectivity of 21 [46].

Ultrathin 30 nm-thick PIM-EA-TB(H₂) composite membranes were fabricated on ultrapermeable PTMSP dense film supports using the Langmuir-Blodgett (LB) technique by Benito *et al.* [47]. The CO₂ permeance

Figure 3



Highlights of PIM research and future membrane development.

and CO₂/N₂ selectivity of the membrane were 114 GPU and 13.5, respectively. A slightly higher CO₂ permeance (134 GPU) and lower CO₂/N₂ selectivity (12.1) were reported by Benito *et al.* [48] for a 30 monolayer PIM-TMN-Trip/PTMSP composite membrane prepared by a similar technique.

Ultrapermeable PIM-1-based thin-film nanocomposite membranes were reported by Bhavsar *et al.* [49], where PIM-1 and hypercrosslinked polystyrene (HCP) nanofiller were blended and dip-coated onto PAN support to form a ~2 µm thick selective layer. The fresh pure PIM-1 TFC membrane showed a high CO₂ permeance of 3333 GPU and CO₂/N₂ selectivity of 14.2 [49]. Dramatic aging effects were reported for these thin-film nanocomposite membranes over a period of 3 months. Nearly 10-fold reduction in CO₂ permeance but with a 73% increase in CO₂/N₂ selectivity from 14.2 to 24.5 was observed for the pure PIM-1 TFC membrane. Interestingly, physical aging was less pronounced in PIM-1/HCP thin-film nanocomposite membranes [49]. Borisov *et al.* [50] successfully prepared a very permeable PIM-1 thin-film composite membranes using a crosslinked PTMSP gutter layer (2.07–3.44 µm) and porous microfiltration membrane support. Fresh, as-cast ultra-thin PIM-1 (0.29–0.42 µm) TFC membranes showed CO₂ permeance in the range of 3010–8010 GPU and CO₂/N₂ selectivity of 35.8–55.7. Similar to other PIM-based TFC membranes, these ultra-thin PIM-1 TFC membranes underwent severe physical aging. The 0.42 µm thick PIM-1 TFC showed nearly 10-fold reduction in CO₂ permeance from 3010 to 281 with a CO₂/N₂ selectivity of 56.2 over 3 months of aging [50].

Liu *et al.* [51] fabricated novel high-performance PIM-1 thin-film nanocomposite membranes using three layers: (1) PIM-1 blended with MOF particles as the selective layer, (2) a modified PDMS gutter layer doped with MOF nanosheets, and (3) a porous polymeric substrate. Pristine PIM-1 TFC membranes showed a CO₂ permeance of 4320 GPU and CO₂/N₂ selectivity of 19. Incorporating MOF-74-Ni and NH₂-UiO-66 nanoparticles into the PIM-1 selective layer of these TFC membranes enhanced both CO₂ permeance to the range of 4660–7460 GPU and CO₂/N₂ selectivity to the range of 26–33 [51]. The pure PIM-1-based TFC membrane experienced rapid aging with 68% loss in CO₂ permeance and 84% increase in CO₂/N₂ selectivity (19 versus 35) after one week and stabilized with a CO₂ permeance of 490 GPU after three weeks. The mixed-matrix PIM-1/MOF TFC membranes revealed stable CO₂ permeance of 900–1200 GPU and CO₂/N₂ selectivity of 26–30 after 4–8 weeks of aging [51]. Future studies focusing on optimizing various TFC membrane fabrication conditions are needed to further improve the gas separation performance of PIM-based TFC membranes.

Another promising alternative way of making PIMs thin-film composite membranes for gas separation is to use the interfacial polymerization reaction, as first demonstrated by Livingston's group for polyarylate membranes using spirobisindane building blocks [52^{*}]. Recently, Ali *et al.* [53^{*}] successfully developed defect-free triptycene-based polyamide TFCs for gas separations through interfacial polymerization by reacting triptycene-acid chloride building blocks with *m*-phenylenediamine (MPD). These polyamide TFCs (MPD) demonstrated excellent salt and organic rejections as well as the potential for H₂/CO₂ separations.

Conclusions

Here, we summarize the important developments of novel ladder PIMs, functionalized PIMs, PIM-derived CMS, and PIM-based thin-film composite membranes for gas separations, especially for some important and challenging energy-intensive applications such as CO₂/N₂, CO₂/CH₄, H₂S/CH₄, N₂/CH₄, and olefin/paraffin separation. Recent advances in PIM materials design has led to a number of breakthroughs which redefined the performance upper bounds for oxygen enrichment (O₂/N₂), hydrogen recovery (H₂/CH₄, H₂/N₂), natural gas treatment (CO₂/CH₄), and carbon capture (CO₂/N₂). However, pure PIM materials are still lacking sufficient gas-pair selectivity, aging and plasticization resistance, and long-term material stability to tackle traditionally difficult olefin/paraffin separations. On the other hand, PIM-derived carbon molecular sieve membranes have shown great promise to significantly improve gas separation properties, especially gas-pair selectivity for these applications. We want to stress that polymer scale-up and reproducible large-scale formation of high-performance

ISA or thin-film composite membranes are of utmost importance to eventually apply this advanced materials class in industrial gas separation processes. Lastly, reproducible methods to control and facilitate physical aging to achieve stable long-term permeation properties of *thin* ladder PIM and PIM-PI membranes are required before acceptance for their use in the gas separation industry.

Conflict of interest statement

Nothing declared.

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Declaration of Competing Interest

The authors report no declarations of interest.

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